

LINACLab: Relativistic Charged-Particle Tracking and Linear Beam Optics from First Principles

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Abstract

LINACLab is a compact and transparent framework for relativistic charged-particle tracking and linear beam optics, designed to emphasize physical clarity and numerical interpretability. Starting from first-principles Lorentz dynamics in $(\mathbf{r}, \mathbf{p})$ phase space, the framework implements explicit time integrators with a clear policy: the relativistic Boris pusher is used by default for magnetic tracking, while fourth-order Runge–Kutta serves as a high-accuracy reference for short or electric-field-dominated studies. Beam-level functionality is provided through transverse second moments, RMS emittance, and Twiss-parameter reconstruction, establishing a direct connection between single-particle tracking and linear optics. Linear transport maps for drifts and quadrupoles are derived directly from the relativistic Hamiltonian under paraxial and fixed-energy approximations, clarifying the assumptions underlying standard matrix optics. Qualitative validation is demonstrated using analytically understood test problems, including helical motion in uniform magnetic fields, kinetic-energy behavior in magnetic tracking, convergence in uniform electric acceleration, and consistency between covariance-based optics and ensemble tracking in simple FODO lattices. A minimal thin-gap RF model illustrates longitudinal energy updates without invoking a full cavity-field description. The framework is intentionally limited: collective effects, self-consistent fields, radiation processes, and full six-dimensional dynamics are excluded by design. Within these bounds, LINACLab serves as a self-contained research and educational tool, suitable for methodological studies, validation of numerical techniques, and as a foundation for future extensions.

1 INTRODUCTION AND SCOPE

Relativistic charged-particle tracking and linear beam optics form the core computational backbone of modern accelerator physics. Accurate numerical integration of particle trajectories, together with a consistent description of beam moments and optical functions, is essential for both conceptual design studies and the interpretation of experimental diagnostics [1–3].

This work presents *LINACLab*, a compact and transparent computational framework for relativistic single-particle and multi-particle beam dynamics. The project is intentionally positioned at the intersection of accelerator physics and computational physics: physical models are derived from first principles, numerical methods are stated explicitly, and implementation choices are kept minimal and auditable.

The scope of the present work is deliberately restricted to:

- relativistic motion of charged particles in prescribed electromagnetic fields,
- numerical integration of the Lorentz equations in $(\mathbf{r}, \mathbf{p})$ phase space,
- transverse beam moments, RMS emittance, and Twiss parameter reconstruction,
- linear beam optics derived from Hamiltonian mechanics under paraxial and fixed-energy approximations.

Several important physical effects are *explicitly excluded* from the current study:

- collective effects such as space charge and wakefields,

- radiation reaction and synchrotron radiation,
- self-consistent field solvers or particle-in-cell methods,
- material interactions and beam–matter coupling.

These exclusions are not limitations of principle, but design choices intended to keep the framework conceptually clear and numerically interpretable.

All equations are formulated in SI units, with normalized accelerator units introduced later where appropriate for beam optics. Validation is qualitative rather than exhaustive: the goal is to demonstrate physical consistency, correct limiting behavior, and algorithmic reliability, rather than to benchmark against large-scale production codes.

The primary intended audience is MSc-by-research and early PhD-level readers with a background in physics or engineering. The exposition therefore emphasizes derivation, physical interpretation, and numerical structure, rather than software performance or feature completeness.

At a Glance

This repository is organized around three notebook-driven threads, with the report providing a consolidated narrative and the figure-generation script producing the report figures deterministically.

- **Notebook 1: Tracking and validation.** Single-particle tracking with explicit integrator comparisons (Boris, Euler, RK4) and analytic-reference tests in uniform fields.
- **Notebook 2: Beam moments and lattice.** Second-moment propagation, Twiss/emittance extraction, and lattice-level demonstrations (FODO matching, mismatch evolution, and a minimal RF energy-gain model).
- **Notebook 3: Hamiltonian.** A compact Hamiltonian formulation leading to linear maps (drift and quadrupole) that connect the physics model to standard accelerator optics.

In the report, the tracking model and qualitative validation are covered first, then the moment-based optics and Twiss formulation, followed by Hamiltonian-based linear maps. The lattice case studies included later in the report mirror the Notebook 2 demonstrations and are presented here in a figure-driven form suitable for standalone reading.

2 RELATIVISTIC FORMULATION IN $(\mathbf{r}, \mathbf{p})$

The fundamental dynamical state evolved in LINACLab is the pair

$$(\mathbf{r}(t), \mathbf{p}(t)), \quad (1)$$

where $\mathbf{r} = (x, y, z)$ is the particle position in meters and $\mathbf{p} = \gamma m \mathbf{v}$ is the mechanical momentum in SI units. This choice is natural for relativistic dynamics because the Lorentz force law is expressed directly in terms of momentum rather than velocity.

The equations of motion are

$$\frac{d\mathbf{r}}{dt} = \mathbf{v}(\mathbf{p}), \quad (2)$$

$$\frac{d\mathbf{p}}{dt} = q [\mathbf{E}(\mathbf{r}, t) + \mathbf{v}(\mathbf{p}) \times \mathbf{B}(\mathbf{r}, t)], \quad (3)$$

with m the particle rest mass and q its charge [4]. Electromagnetic fields are treated as externally prescribed functions of space and time.

2.1 Velocity reconstruction from momentum

In relativistic dynamics, momentum and velocity are related through the Lorentz factor γ ,

$$\mathbf{p} = \gamma m \mathbf{v}, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}. \quad (4)$$

Rather than solving implicitly for $\mathbf{v}$, it is more robust to express γ directly as a function of $\mathbf{p}$. Starting from the energy–momentum relation [4],

$$E^2 = (pc)^2 + (mc^2)^2, \quad (5)$$

and identifying $E = \gamma mc^2$, one obtains

$$\gamma(\mathbf{p}) = \sqrt{1 + \frac{\mathbf{p} \cdot \mathbf{p}}{m^2 c^2}}. \quad (6)$$

The velocity follows immediately as

$$\mathbf{v}(\mathbf{p}) = \frac{\mathbf{p}}{\gamma(\mathbf{p}) m}. \quad (7)$$

This formulation avoids iterative inversion and is numerically stable for both single-particle and multi-particle tracking. In the implementation, this mapping is vectorized so that the same functions apply to individual particles and ensembles without modification.

2.2 Sign conventions and physical interpretation

For electrons, the charge is $q = -e$. Consequently, an electric field pointing in the $+z$ direction produces a force in the $-z$ direction. Care is taken throughout the implementation to maintain consistent sign conventions between analytical expressions and numerical results.

The $(\mathbf{r}, \mathbf{p})$ formulation also clarifies several important physical limits:

- In purely electric fields, changes in $|\mathbf{p}|$ correspond directly to changes in kinetic energy.
- In purely magnetic fields, the force is perpendicular to $\mathbf{v}$, implying exact conservation of γ in the continuous equations.
- Numerical violation of this conservation therefore provides a sensitive diagnostic of integrator behavior.

These properties motivate both the choice of evolved variables and the integrator policy adopted in the following sections.

3 NUMERICAL INTEGRATION AND STRUCTURE

3.1 Baseline Integrators: Euler and RK4

The Lorentz equations in momentum form constitute a coupled system of ordinary differential equations,

$$\frac{d\mathbf{r}}{dt} = \mathbf{v}(\mathbf{p}), \quad (8)$$

$$\frac{d\mathbf{p}}{dt} = q [\mathbf{E}(\mathbf{r}, t) + \mathbf{v}(\mathbf{p}) \times \mathbf{B}(\mathbf{r}, t)], \quad (9)$$

where the velocity is obtained from the relativistic energy–momentum relation.

Two standard explicit integrators are implemented as baselines: the forward Euler method and the classical fourth-order Runge–Kutta (RK4) method [5]. Euler integration is included primarily for pedagogical purposes and as a low-order reference; it is not suitable for accurate long-term tracking.

RK4 provides high local accuracy and is well suited for short integrations or problems dominated by electric fields. However, RK4 does not preserve the geometric structure of Lorentz motion. In purely magnetic fields, where the exact dynamics correspond to a rotation in momentum space at constant kinetic energy, RK4 can exhibit slow but systematic energy drift over long integration times. This behavior reflects the non-symplectic nature of generic Runge–Kutta methods [6] rather than a flaw in the implementation.

Accordingly, RK4 is treated in this work as a *reference integrator* for benchmarking and controlled short-time studies, not as a default method for magnetic tracking.

3.2 Relativistic Boris Pusher

For problems involving magnetic fields, the primary integrator adopted in LINACLab is the relativistic Boris pusher [7–9]. The Boris method splits the momentum update over a time step Δt into a sequence of substeps:

1. a half-step electric “kick”,
2. a rotation in momentum space due to the magnetic field,
3. a second half-step electric “kick”,

followed by a position update using the new momentum.

The key property of the method is that the magnetic substep corresponds to a pure rotation, which preserves the magnitude of the momentum vector. As a result, in the absence of electric fields, the Boris method exhibits excellent long-term behavior with respect to kinetic-energy conservation.

In the relativistic formulation, the rotation strength depends on the Lorentz factor γ . In this implementation, γ is evaluated using the intermediate momentum after the first half electric kick, a standard and physically consistent choice for relativistic tracking.

Based on both theoretical considerations and qualitative validation tests, the following *integrator policy* is adopted throughout this work:

- the relativistic Boris pusher is the default choice for magnetic elements and long tracking runs,
- RK4 is used as a high-accuracy reference method for short integrations or electric-field-dominated problems,
- Euler integration is retained only as a low-order baseline.

This policy reflects the physical structure of the underlying equations rather than purely numerical order considerations.

4 QUALITATIVE VALIDATION AND WHAT IS ESTABLISHED

The purpose of the validation studies presented in this work is not exhaustive benchmarking against established accelerator codes, but rather to verify that the implemented equations and numerical methods reproduce the correct physical behavior in well-understood limiting cases. All validation is therefore qualitative in nature and focused on invariants, symmetries, and characteristic motion.

4.1 Uniform Magnetic Field: Helical Motion

In the absence of electric fields, the Lorentz force reduces to

$$\frac{d\mathbf{p}}{dt} = q \mathbf{v} \times \mathbf{B}, \quad (10)$$

which is always perpendicular to the velocity. Consequently, the kinetic energy and Lorentz factor γ are conserved exactly in the continuous equations of motion.

For a uniform magnetic field $\mathbf{B} = B_z \hat{\mathbf{z}}$, the exact solution is a helix: uniform circular motion in the transverse plane combined with uniform drift along the field direction. Figure 1 shows the transverse projection of such a trajectory obtained using the relativistic Boris pusher, while Figure 2 shows the corresponding longitudinal motion.

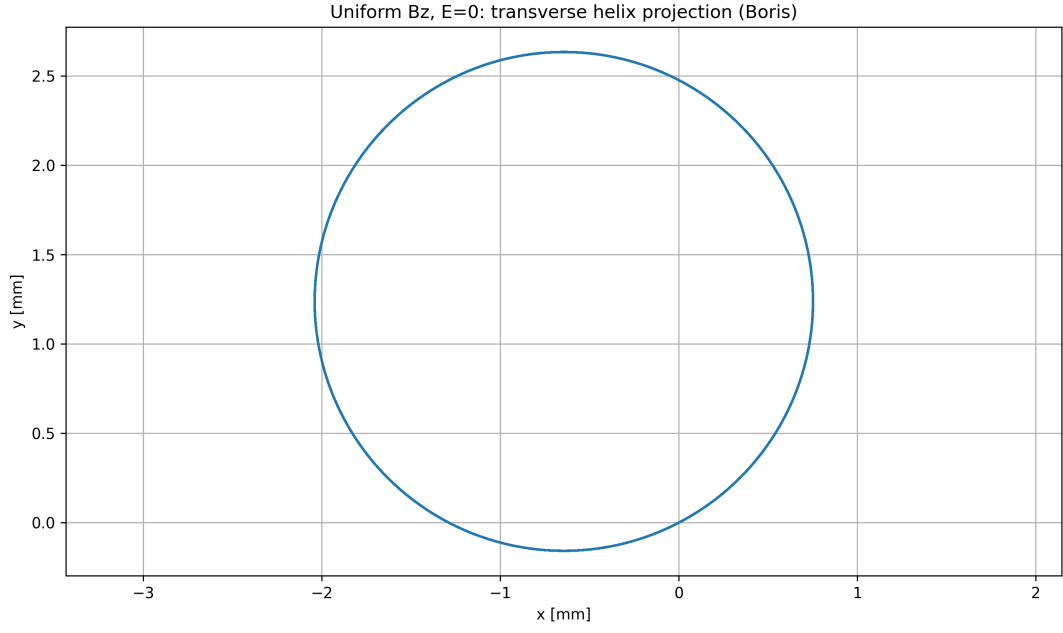


Figure 1: Uniform B_z and $E = 0$: transverse helical projection from relativistic Boris integration.

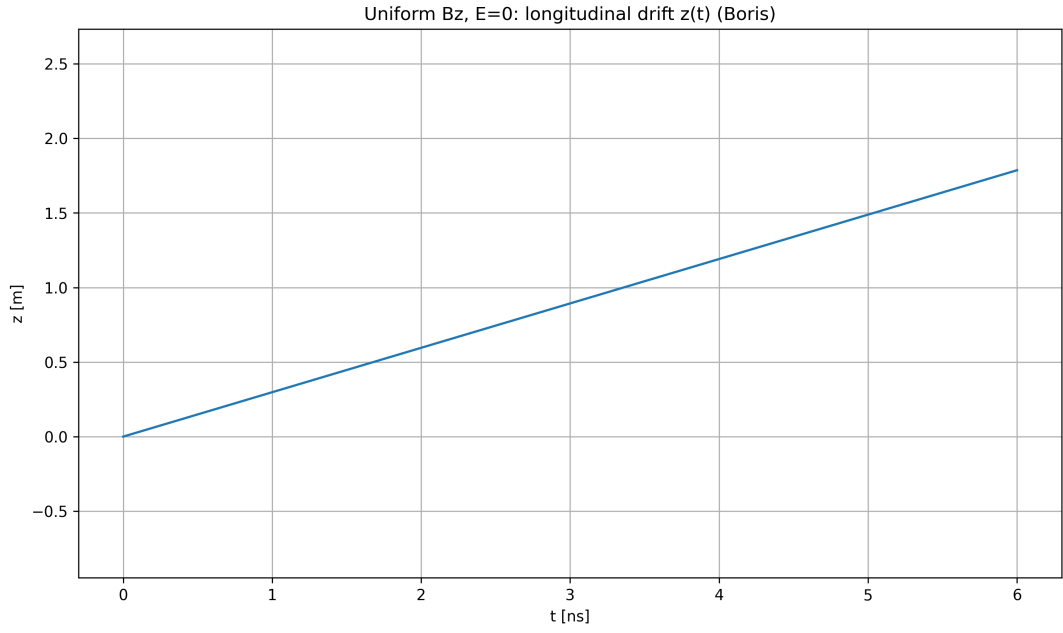


Figure 2: Uniform B_z and $E = 0$: longitudinal drift $z(t)$ from relativistic Boris integration.

The numerical results exhibit the expected circular transverse motion and linear longitudinal drift, confirming correct coupling between momentum rotation and velocity reconstruction. No artificial spiraling or secular distortion is observed over the integration interval.

4.2 Uniform E_z benchmark (analytic reference)

The uniform- B helix tests primarily validate magnetic-field handling and qualitative long-time stability, but do not directly test the electric-field update in a regime where the dynamics are E -dominated. To isolate that case, we consider a uniform longitudinal field $E_z = \text{const}$ with zero magnetic field and compare numerical tracking against an analytic reference.

For an electron starting from rest with motion constrained along z , the canonical momentum evolves as $p_z(t) = qE_z t$ and the Lorentz factor becomes $\gamma(t) = \sqrt{1 + (p_z/mc)^2}$. The longitudinal position follows

$$z(t) = \frac{mc^2}{qE_z} (\gamma(t) - 1), \quad (11)$$

which provides a closed-form final-time reference for measuring discretization error.

Figure 3 reports the final-time position error as a function of timestep. In this regime, RK4 exhibits the expected near-fourth-order behavior over the pre-asymptotic range, while the Boris pusher remains stable but is not designed for high-order accuracy in purely electric acceleration. This benchmark complements the uniform- B tests by explicitly validating the electric-field update against a known analytic trajectory.

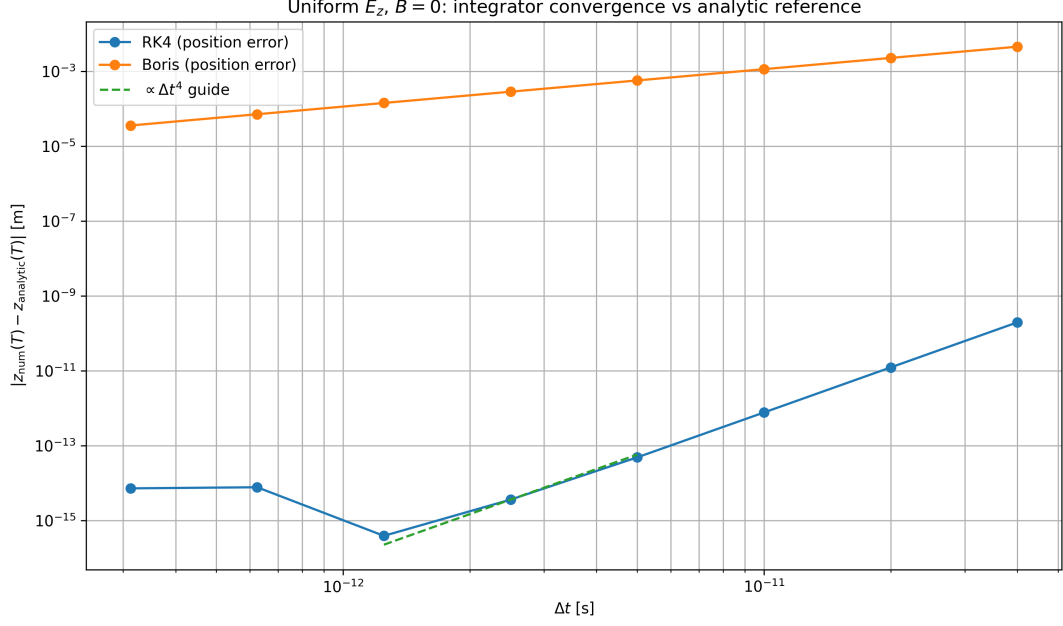


Figure 3: Final-time position error in a uniform E_z field versus timestep. The analytic reference follows $p_z(t) = qE_z t$ and $z(t) = \frac{mc^2}{qE_z}(\gamma(t) - 1)$.

4.3 Energy Behavior in Magnetic Tracking

Although energy is conserved analytically for purely magnetic motion, numerical integrators may violate this invariant. Figure 4 compares the relative kinetic-energy change obtained using RK4 and the relativistic Boris pusher in a uniform magnetic field.

4.4 Energy Behavior in Magnetic Tracking

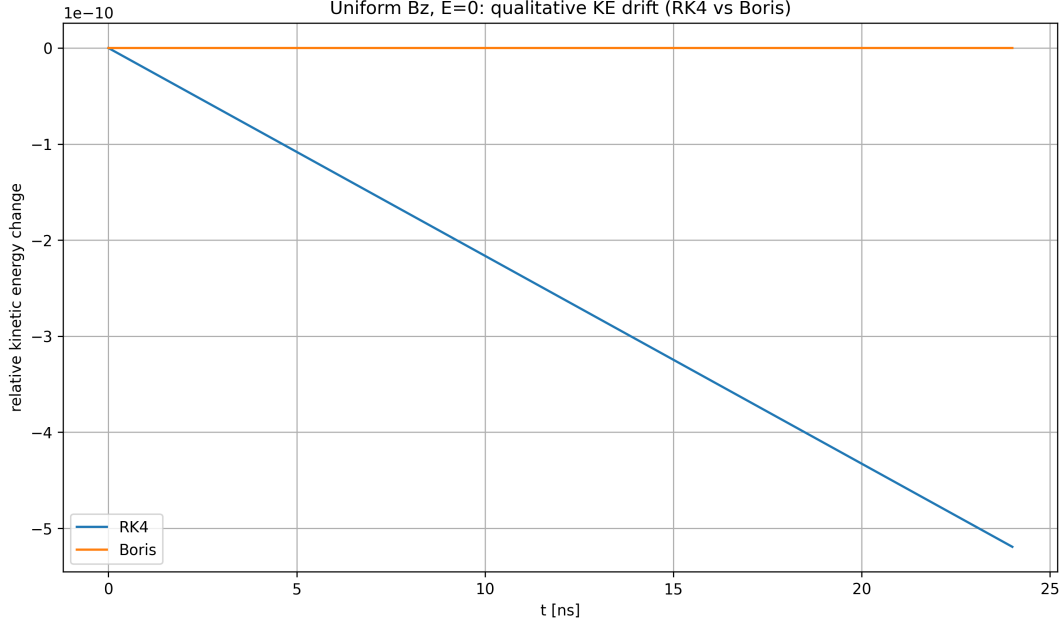


Figure 4: Qualitative kinetic-energy drift in uniform B_z with $E = 0$: RK4 vs Boris.

As expected, RK4 exhibits a slow secular drift in kinetic energy over long integration times, despite its high formal order. In contrast, the Boris pusher maintains near-constant kinetic energy, with only small bounded oscillations. This qualitative behavior reflects the geometric structure of the Boris method and motivates its use as the default integrator for magnetic elements.

4.5 What Is and Is Not Established

These tests establish the following:

- correct implementation of the relativistic Lorentz equations in $(\mathbf{r}, \mathbf{p})$ form,
- correct reconstruction of velocity from momentum,
- physically consistent behavior of the Boris pusher in magnetic fields,
- qualitative agreement with known analytic solutions.

The following are *not* established by the present validation:

- long-term quantitative error bounds,
- performance comparisons with large-scale accelerator codes,
- accuracy in the presence of collective or self-consistent effects.

The validation scope is therefore aligned with the intended use of the framework as a transparent research and educational tool rather than a production-grade tracking engine.

5 TRANSVERSE BEAM MOMENTS, EMITTANCE, AND TWISS PARAMETERS

Beyond single-particle tracking, accelerator physics relies heavily on the second-moment description of particle ensembles. LINACLab implements a transverse-only (4D) beam formalism that bridges statistical beam optics and full phase-space tracking.

5.1 Transverse Phase-Space Coordinates

In the paraxial approximation, the transverse phase-space coordinates are defined as [1, 2]

$$\mathbf{u} = (x, x', y, y')^T, \quad (12)$$

where x and y are transverse displacements and the slopes are defined by

$$x' \equiv \frac{dx}{dz}, \quad y' \equiv \frac{dy}{dz}. \quad (13)$$

For small angles, these slopes are related to mechanical momentum by

$$x' \approx \frac{p_x}{p_z}, \quad y' \approx \frac{p_y}{p_z}. \quad (14)$$

This approximation provides a direct bridge between linear optics and the $(\mathbf{r}, \mathbf{p})$ tracking formulation.

5.2 Second Moments and RMS Emittance

Given an ensemble of N macroparticles $\{\mathbf{u}_i\}$, the mean and covariance matrix are defined as

$$\bar{\mathbf{u}} = \langle \mathbf{u} \rangle, \quad (15)$$

$$\Sigma = \langle (\mathbf{u} - \bar{\mathbf{u}})(\mathbf{u} - \bar{\mathbf{u}})^T \rangle. \quad (16)$$

For a single transverse plane, e.g. (x, x') , the RMS (geometric) emittance is defined by [1, 2, 10]

$$\varepsilon_x = \sqrt{\det \Sigma_x}, \quad (17)$$

where Σ_x is the 2×2 covariance submatrix. The corresponding Twiss parameters are obtained as

$$\beta_x = \frac{\Sigma_{11}}{\varepsilon_x}, \quad \alpha_x = -\frac{\Sigma_{12}}{\varepsilon_x}, \quad \gamma_x = \frac{\Sigma_{22}}{\varepsilon_x}, \quad (18)$$

which satisfy the identity $\beta_x \gamma_x - \alpha_x^2 = 1$.

5.3 Beam Generation and Consistency

Both Gaussian and waterbag transverse distributions are supported. Beam generation is performed directly from target emittance and Twiss parameters, and the resulting sample covariance is verified to reproduce the intended second moments within statistical fluctuations. Figure 5 shows a representative transverse phase-space projection for a Gaussian beam. The recovered emittance and Twiss parameters are consistent with the input values, confirming the internal consistency of the moment-based formulation.

5.4 Mapping to Tracking Variables

To initialize tracking simulations, the transverse beam coordinates are mapped to full phase-space variables via

$$p_x = x' p_{z,\text{ref}}, \quad p_y = y' p_{z,\text{ref}}, \quad p_z = p_{z,\text{ref}}, \quad (19)$$

with $p_{z,\text{ref}}$ obtained from a chosen reference kinetic energy. This mapping provides a controlled and explicit connection between linear optics and relativistic tracking.

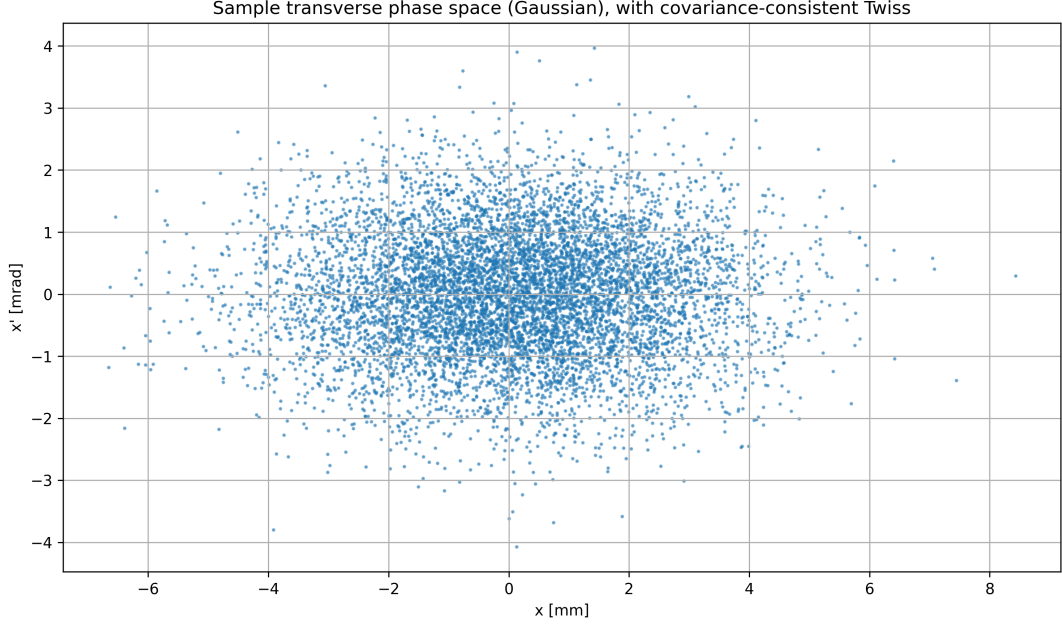


Figure 5: Example transverse phase space projection $x-x'$ from a Gaussian beam consistent with target second moments.

6 LATTICE CASE STUDIES (NOTEBOOK 2 DEMONSTRATIONS)

The previous section establishes the moment-based optics machinery (second moments, emittance, and Twiss parameters). This section applies those tools to compact lattice demonstrations that connect the formalism to standard accelerator-design workflows. The goal is not to claim a complete linac design model, but to show that (i) the matrix optics and Twiss-based predictions behave correctly in periodic lattices, and (ii) moment-based tracking remains consistent with the corresponding linear-optics propagation under the hard-edge assumptions adopted throughout this project.

6.1 Matched periodic FODO envelope

We first consider a simple periodic FODO cell constructed from hard-edge quadrupoles and drifts. From the one-period transfer matrix, matched Twiss parameters are obtained and used to build the initial second-moment matrix for a chosen normalized emittance. The envelope is then propagated along the lattice using $\Sigma(s) = M(s)\Sigma_0 M(s)^\top$, where $M(s)$ is the accumulated transfer matrix along the longitudinal coordinate.

Figure 6 shows the resulting matched transverse envelope. The periodic “breathing” of the envelope is the expected behavior of a stable focusing channel, and serves as a basic sanity check that the lattice assembly, transfer matrices, and Twiss reconstruction are internally consistent under the linear model.

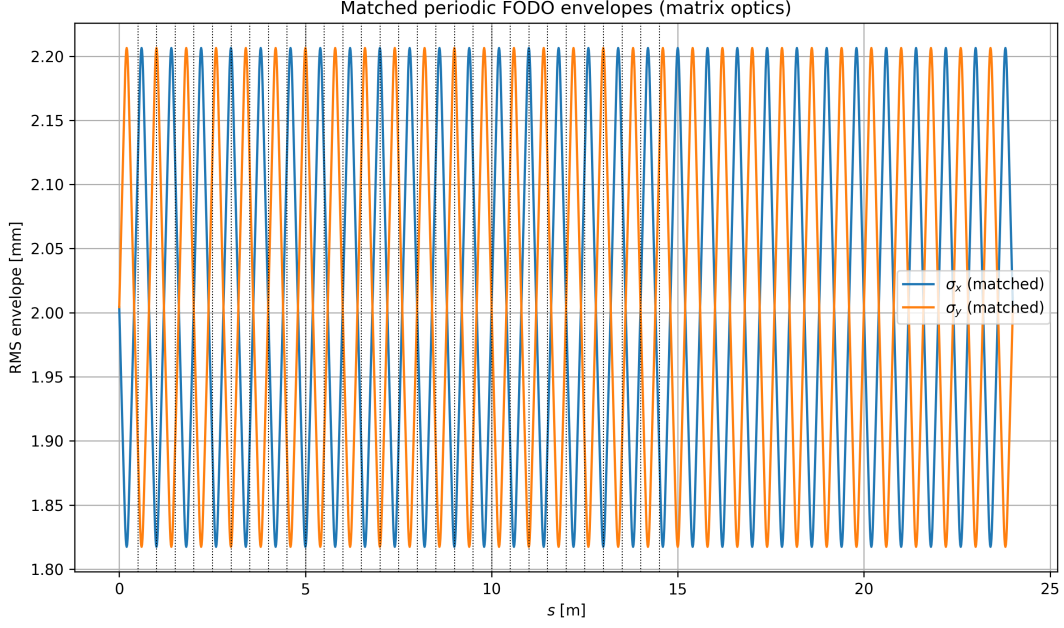


Figure 6: Matched transverse envelope in a periodic FODO lattice obtained from the one-period matrix and propagated using $\Sigma(s) = M(s)\Sigma_0 M(s)^\top$.

6.2 Short-FODO mismatch evolution: tracking versus matrix optics

A second test compares the envelope evolution predicted by matrix propagation against a moment-based tracking calculation over a short FODO section. The tracking route advances an ensemble under the same hard-edge field model, while the matrix route propagates second moments via transfer matrices. For small amplitudes and under hard-edge assumptions, both approaches should agree up to statistical noise and discretization choices.

Figure 7 compares RMS beam envelopes in a short FODO lattice for intentionally mismatched initial Twiss parameters. Solid curves show envelopes obtained from linear matrix optics via covariance propagation, while dashed curves show envelopes extracted from particle tracking in hard-edge quadrupole fields. Both approaches predict bounded envelope oscillations with slow beating, characteristic of mismatch in a periodic focusing channel. The close agreement between matrix and tracking results confirms the consistency of the lattice model, beam initialization, and tracking implementation in the linear regime.

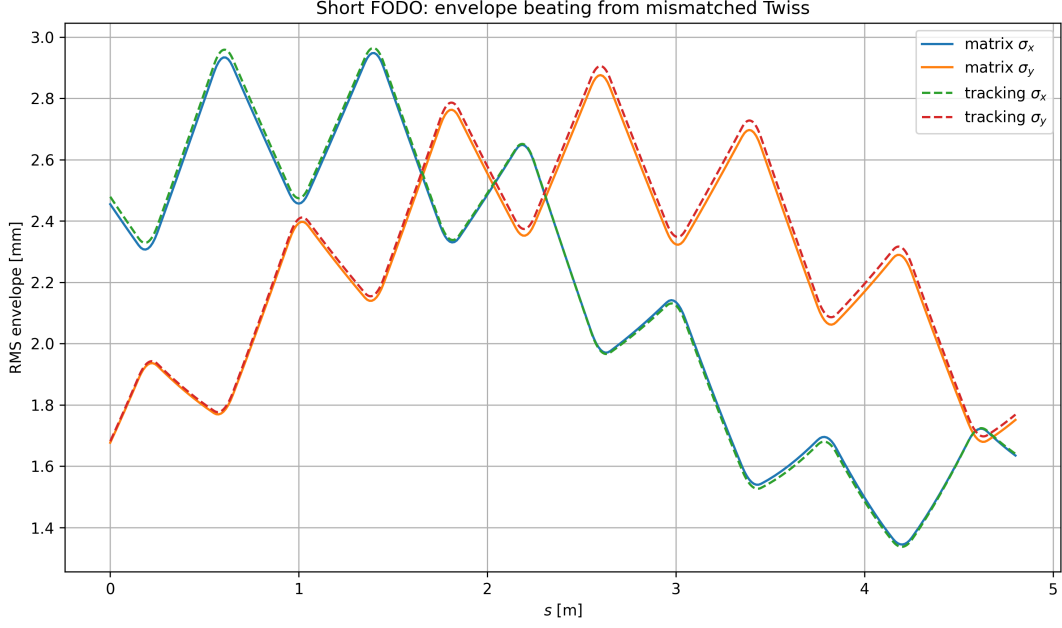


Figure 7: Short-FODO envelope comparison using (i) moment-based tracking of an ensemble under the hard-edge model and (ii) transfer-matrix propagation of $\Sigma(s)$. The plotted mismatch metric quantifies deviations in second moments.

6.3 Minimal RF model: thin-gap energy gain

Finally, we include a minimal longitudinal extension in which acceleration is represented as a thin-gap energy kick, separated by drifts. This is intentionally limited: it is not a self-consistent cavity-field model and does not include phase stability, transit-time factors, or detailed longitudinal dynamics. Its purpose is to demonstrate that the code path can couple transverse tracking with a simple energy-update mechanism consistent with the stated project scope.

Figure 8 shows the energy gain accumulated across a sequence of thin gaps. The energy exhibits periodic modulation as a function of longitudinal coordinate due to RF phase advance between gaps. This minimal model is not intended to represent a full accelerating structure, but rather to validate the RF-kick implementation and its integration with the beam dynamics framework.

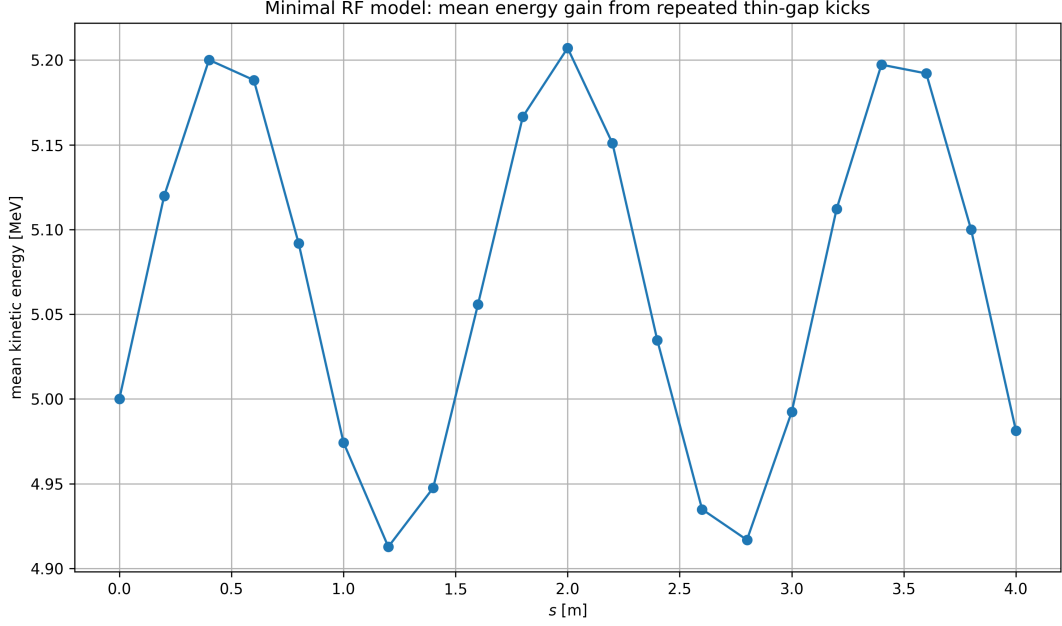


Figure 8: Minimal RF acceleration model implemented as repeated thin-gap energy kicks separated by drifts. This demonstration is intended as a bookkeeping and integration test rather than a full longitudinal dynamics model.

7 HAMILTONIAN LINEAR OPTICS AND EXACT MAPS

Linear beam optics is most naturally derived from Hamiltonian mechanics [1, 11]. In this section, transverse dynamics is obtained directly from the relativistic Hamiltonian, without invoking transfer matrices as a starting assumption.

7.1 Relativistic Hamiltonian and Choice of Independent Variable

For a charged particle in electromagnetic fields, the Hamiltonian with time as the independent variable is [4]

$$H_t(\mathbf{r}, \mathbf{p}) = \sqrt{(\mathbf{p} - q\mathbf{A})^2 c^2 + m^2 c^4} + q\Phi. \quad (20)$$

We restrict attention to static magnetic fields with $\Phi = 0$ and $\partial\mathbf{A}/\partial t = 0$, so that H_t is conserved.

In accelerator physics, it is convenient to use the longitudinal coordinate z as the independent variable. This is achieved through a canonical transformation that exchanges the roles of t and z , leading to an effective Hamiltonian

$$H = -p_z. \quad (21)$$

7.2 Paraxial and Fixed-Energy Approximation

Assuming a fixed reference momentum p_0 and paraxial motion $|p_x|, |p_y| \ll p_0$, the longitudinal momentum can be expanded to second order in the transverse variables. Dropping constant terms yields an effective transverse Hamiltonian of the form

$$H = \frac{p_x^2 + p_y^2}{2p_0} - \frac{p_0 k(z)}{2} (x^2 - y^2), \quad (22)$$

where $k(z)$ is the quadrupole focusing strength.

This Hamiltonian is quadratic and separable, leading to linear equations of motion in each transverse plane.

7.3 Exact Drift and Quadrupole Maps

For a drift space ($k = 0$), the equations of motion integrate exactly to

$$\begin{pmatrix} x \\ p_x \end{pmatrix}_L = \begin{pmatrix} 1 & L/p_0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ p_x \end{pmatrix}_0, \quad (23)$$

with an analogous expression in the y -plane.

For a hard-edge quadrupole with constant k , the equations reduce to harmonic or hyperbolic motion depending on the sign of k . The resulting transport maps are exact and symplectic, and correspond to the standard focusing and defocusing matrices used in linear optics.

Importantly, these maps are not postulated but derived directly from the Hamiltonian, clarifying the assumptions and limits under which matrix optics is valid.

8 LIMITATIONS AND FUTURE WORK

8.1 Limitations

LINACLab is intentionally restricted in scope, and several limitations follow directly from its design philosophy. Electromagnetic fields are treated as externally prescribed; there is no self-consistent treatment of collective effects such as space charge, wakefields, or beam loading, nor any coupling to field solvers or particle-in-cell methods. As a result, the framework is not suitable for high-intensity or collective-dominated regimes.

The beam model is transverse-only (4D) for optics and moment-based studies. Longitudinal dynamics are represented only in a minimal sense through thin-gap energy kicks, without phase stability, transit-time factors, synchrotron motion, or energy-spread evolution. The mapping between transverse slopes and mechanical momentum assumes a fixed reference longitudinal momentum and is therefore valid only under paraxial and near-isoenergetic conditions.

All lattice elements are modeled using idealized hard-edge fields. Fringe fields, misalignments, higher-order multipoles, and realistic field maps are not included. While this is sufficient for linear-optics demonstrations and consistency checks, it limits quantitative applicability to real accelerator components.

Finally, validation throughout this work is qualitative rather than exhaustive. The numerical methods are tested against analytically understood limits and internal consistency criteria, but no systematic benchmarking against established production accelerator codes is performed. Consequently, the framework should be viewed as a transparent research and educational tool rather than a validated design or simulation engine.

8.2 Future Work

Several extensions follow naturally from the present framework. A primary direction is a full six-dimensional formulation with explicit longitudinal phase and energy deviation, enabling self-consistent RF cavities beyond the thin-gap approximation and allowing phase stability and synchrotron motion to be studied within the same formalism.

On the optics side, the element set can be expanded to include solenoids, sector bends, and controlled fringe-field models, while preserving the Hamiltonian-based derivation of linear maps. This would broaden the range of lattice configurations accessible without sacrificing physical clarity.

From a numerical perspective, additional structure-preserving schemes for combined electric and magnetic fields can be introduced, while retaining the current integrator policy in which the relativistic Boris pusher is used by default for magnetic tracking and RK4 serves as a high-accuracy reference. Finally, the diagnostics layer can be extended toward inference-oriented tools, such as quadrupole-scan emittance reconstruction with uncertainty propagation and regression tests that enforce physical invariants as the code evolves.

9 CONCLUSION

This work presented LINACLab, a compact and transparent framework for relativistic charged-particle tracking and linear beam optics. Starting from first-principles dynamics in $(\mathbf{r}, \mathbf{p})$ phase space, the framework combines physically motivated numerical integration with moment-based beam diagnostics and Hamiltonian-derived linear optics.

Qualitative validation against analytically understood test cases demonstrated correct physical behavior in key limits. In particular, the relativistic Boris pusher was shown to preserve kinetic energy in purely magnetic motion, while RK4 provided high local accuracy for short integrations and electric-field-dominated problems. Transverse beam moments, RMS emittance, and Twiss parameters were recovered consistently from tracked ensembles, establishing a clear bridge between single-particle dynamics and linear optics.

Lattice case studies based on periodic and mismatched FODO channels confirmed that covariance-based matrix propagation and moment-extracted tracking results agree in the linear regime under hard-edge assumptions. A minimal RF model further demonstrated that longitudinal energy updates can be integrated consistently with transverse dynamics, without overextending the physical scope of the framework.

Linear transport maps were derived directly from the relativistic Hamiltonian under controlled paraxial and fixed-energy approximations, clarifying the assumptions underlying standard matrix optics and emphasizing the role of canonical structure.

Within its deliberately limited scope, LINACLab provides a physically transparent, numerically interpretable foundation for studying relativistic beam dynamics. It is intended as a research-oriented technical note and a basis for future extensions rather than as a production accelerator code.

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A UNITS AND CONVENTIONS

All primary equations of motion in LINACLab are formulated in SI units. This appendix summarizes the conventions used throughout the paper and the implementation.

A.1 Physical Units

The fundamental quantities are expressed as:

- position $\mathbf{r}$ in meters (m),
- time t in seconds (s),
- mechanical momentum $\mathbf{p}$ in kg m s^{-1} ,
- electric field $\mathbf{E}$ in V m^{-1} ,
- magnetic field $\mathbf{B}$ in tesla (T),
- mass m in kilograms (kg),
- charge q in coulombs (C).

Derived quantities follow standard relativistic definitions:

$$\gamma = \sqrt{1 + \frac{\mathbf{p} \cdot \mathbf{p}}{m^2 c^2}}, \quad \mathbf{v} = \frac{\mathbf{p}}{\gamma m}. \quad (24)$$

Kinetic energy is computed as

$$K = (\gamma - 1)mc^2, \quad (25)$$

and is reported in joules internally, with electronvolts used only for presentation where convenient.

A.2 Coordinate System and Sign Conventions

A right-handed Cartesian coordinate system (x, y, z) is used throughout. The longitudinal direction is aligned with the nominal beam axis and taken to be the $+z$ direction.

For electrons, the charge is $q = -e$. As a result:

- an electric field in the $+z$ direction accelerates electrons toward $-z$,
- magnetic forces follow the right-hand rule applied to $\mathbf{v} \times \mathbf{B}$ with the electron charge included explicitly.

These sign conventions are maintained consistently between analytical expressions, numerical integration, and plotted results.

A.3 Paraxial Variables and Normalized Quantities

In linear optics and beam diagnostics, transverse slopes are defined as

$$x' \equiv \frac{dx}{dz}, \quad y' \equiv \frac{dy}{dz}. \quad (26)$$

Under the paraxial approximation, these are related to momentum components by

$$x' \approx \frac{p_x}{p_z}, \quad y' \approx \frac{p_y}{p_z}, \quad (27)$$

with p_z evaluated using a fixed reference momentum in the absence of acceleration.

Normalized emittance is defined in the usual way,

$$\varepsilon_n = \beta \gamma \varepsilon, \quad (28)$$

where ε is the geometric RMS emittance and $\beta = v/c$.

B API DOCUMENTATION

This appendix summarizes the public application programming interface (API) of LINACLab. The implementation is intentionally contained within a single source file, `linacclab.py`, to emphasize conceptual clarity and ease of inspection. Functions are grouped by physical role rather than by implementation order.

B.1 Physical Constants and Unit Conversions

Fundamental physical constants are defined explicitly in SI units. These include the speed of light c , the elementary charge e , and particle rest masses. Convenience functions are provided for unit conversion, including electronvolt to joule and the inverse.

These definitions ensure that all numerical calculations remain dimensionally consistent and transparent.

B.2 Particle Representation

Particles are represented by lightweight data structures containing mass and charge. A predefined electron instance is provided for convenience. This design allows the same tracking infrastructure to be reused for different particle species without modification of the integrators or field models.

B.3 Relativistic Kinematics

A set of helper functions implements relativistic kinematic relations. These include:

- computation of the Lorentz factor γ from momentum,
- computation of velocity from momentum and mass,
- conversion between kinetic energy and momentum.

All relations are derived directly from the relativistic energy–momentum identity and are implemented in vectorized form to support ensemble tracking.

B.4 Electromagnetic Field Models

Electromagnetic fields are represented as callable functions returning field vectors at a given position and time. Simple factory functions are provided for common test cases, including uniform electric and magnetic fields.

This functional representation allows arbitrary external field configurations to be introduced without altering the tracking logic.

B.5 Equations of Motion

The right-hand side of the Lorentz equations is implemented as a dedicated function that computes time derivatives of position and momentum. This separation enables the same equations of motion to be integrated by multiple numerical schemes and simplifies verification and testing.

B.6 Time Integration Schemes

Three explicit time-integration schemes are provided:

- forward Euler integration,
- classical fourth-order Runge–Kutta (RK4),
- the relativistic Boris pusher.

Each integrator advances the state $(\mathbf{r}, \mathbf{p})$ over a single time step. The Boris pusher is implemented in a structure-preserving form suitable for long-term magnetic tracking, while RK4 serves as a high-accuracy reference method.

B.7 Tracking Wrappers

High-level wrapper functions are provided for tracking single particles and ensembles over multiple time steps. These functions manage time stepping, data storage, and integrator selection, and return complete trajectory histories for post-processing.

This layer separates numerical integration from data management and plotting.

B.8 Beam Generation and Statistics

Functions are provided to generate transverse particle ensembles with prescribed RMS emittance and Twiss parameters. Both Gaussian and waterbag distributions are supported.

Additional routines compute first and second moments of the beam distribution, including RMS emittance and Twiss parameters recovered from the covariance matrix. These routines establish the connection between statistical beam optics and full phase-space tracking.

B.9 Linear Optics and Transport Maps

Exact linear transport maps for drift spaces and quadrupole elements are implemented based on the Hamiltonian formulation presented in the main text. These maps operate directly on transverse phase-space coordinates and preserve symplectic structure by construction.

They are used both for standalone optics calculations and for comparison with tracking-based results.

B.10 Diagnostics and Derived Quantities

A small set of helper routines computes derived quantities such as kinetic energy, normalized emittance, and beam envelopes. These functions are used for validation and visualization rather than as part of the core integration loop.

B.11 Design Philosophy

The API is intentionally minimal. No global state is maintained, and all functions operate on explicitly provided inputs. This design facilitates reproducibility, testing, and extension, and reflects the intended use of LINACLab as a research and educational framework rather than a production accelerator code.

C MINIMAL REPRODUCIBILITY CHECKLIST

This appendix summarizes the minimal steps required to reproduce the results and figures presented in this report using the public LINACLab repository. The framework is designed to require only a standard Python environment and commonly used scientific libraries.

C.1 Software Environment

The following environment was used for all results presented in this work:

- Python ≥ 3.10 ,
- NumPy ≥ 1.26 ,
- Matplotlib ≥ 3.7 ,
- JupyterLab ≥ 4.0 ,
- IPyKernel ≥ 6.0 ,
- a standard L^AT_EX distribution (e.g. TeX Live) with `latexmk`.

Exact package versions are listed in the repository file `requirements.txt`.

C.2 Repository Structure

The repository is organized as follows:

- `src/`: core implementation (`linaclab.py`),
- `notebooks/`: theory, validation, and application notebooks,

- `report/`: L^AT_EX source, figures, and build scripts,
- `figures/`: figures generated from notebooks.

C.3 Reproducing Report Figures

All figures appearing in this report can be regenerated deterministically using the script:

```
python report/scripts/make_figures.py
```

This script writes all report-specific figures to `report/figures/`.

C.4 Building the Report

To build the report PDF, run:

```
cd report
./build.sh
```

This command assumes that all required figures are already present in `report/figures/`. A clean rebuild can be performed using:

```
./clean.sh
./build.sh
```

C.5 Notebook Execution Order

For readers interested in the full derivation and exploratory workflow, the notebooks should be executed in the following order:

1. `01_tracking_and_validation.ipynb`,
2. `02_beam_and_lattice.ipynb`,
3. `03_hamiltonian.ipynb`.

The notebooks are not required to reproduce the report figures but provide additional derivations, validation tests, and interpretive context.